IDEATION PHASE

EV Market Analysis Dashboard using Tableau Public

Dataset-based Empathy Map, Problem Statements, Brainstorming & Idea Prioritization

Dataset snapshot: 200,000 rows × 37 columns. The dataset contains EV specifications, sales, pricing, battery information, ratings, manufacturing geography, and recommended use cases.

BRAINSTORM & IDEA PRIORITIZATION (4 Marks)

Project Details	Information
Date	18 October 2026
Team ID	Roll 58
Project Name	EV Market Analysis Dashboard using Tableau Public
Primary Users	EV market strategy director, automotive business intelligence manager, sustainable mobility researcher, students and project reviewers
Purpose	Understand what users need to compare EV market sales and vehicle specifications clearly.

Step 1: Brainstorm and idea listing. Ideas are listed first without judging them, then grouped by the main user need.

S.No.	Team Member	Idea / Suggestion	Category	Group / No.
1	Team Member 1	Create a dashboard to compare annual sales by manufacturer and vehicle category.	Sales Analytics	1
2	Team Member 2	Show battery-type distribution and compare battery capacity, cycle life and degradation.	Battery Technology	2
3	Team Member 3	Compare range_km, efficiency_wh_per_km and charging performance across categories.	Range & Efficiency	3
4	Team Member 4	Explore price_usd against range_km to compare vehicle price and driving range.	Pricing Analysis	4

5	Team Member 5	Compare customer_rating and safety_rating_stars by manufacturer or category.	Customer & Safety Insights	5
6	Team Member 6	Explore annual sales by manufacturing_country, market_region and model_year.	Geographic & Time Analysis	6
7	Team Member 7	Compare co2_saved_tons_per_year and recommended_use_case across EV categories.	Sustainability & Use Case	7

Step 2: Idea prioritization. Feasibility reflects whether the idea can be built using the supplied columns and Tableau Public. Importance reflects its usefulness to the primary users.

Group / No.	Final Idea	Feasibility	Importance	Priority	Selected
1	Interactive EV Market Analysis Dashboard with KPI cards, manufacturer/category sales, battery distribution, range, price and customer-rating views.	High	High	1	Yes
2	EV battery technology and performance comparison using battery type, capacity, cycle life, degradation, range and charging fields.	High	High	2	Yes — as dashboard views
3	Price-versus-range analysis to help users explore value comparisons between EV models.	High	High	3	Yes — as dashboard view
4	Geographic and model-year analysis of recorded sales by manufacturing country and market region.	High	Medium	4	Yes — as dashboard view
5	Safety, customer rating and sustainability comparison by vehicle category/manufacturer.	Medium	Medium	5	Optional, if space and data quality allow

6	Predict future EV sales using machine learning.	Medium	Medium	6	No — outside the current descriptive-analysis scope

Why This Idea Was Selected

- High feasibility: the required fields are already available in the supplied dataset and can be analyzed using Tableau Public.
- High importance: users can compare many EV dimensions in one place instead of manually reviewing a large spreadsheet.
- Clear deliverable: an interactive dashboard is suitable for a class project, demonstration and review.
- Manageable scope: descriptive analytics is achievable without advanced machine-learning training.
- Responsible interpretation: the dashboard will describe patterns in the supplied dataset and avoid claiming that correlation proves causation.

Dataset Fields to Use

Analysis Area	Relevant Dataset Fields	Suggested Visualization
Sales overview	annual_sales_units, manufacturer, category	KPI cards; bar chart by manufacturer/category
Battery technology	battery_type, battery_capacity_kwh, battery_cycle_life, battery_degradation_pct_5yr	Bar chart; category comparison
Range and efficiency	range_km, efficiency_wh_per_km, category	Bar chart; scatter plot
Pricing and charging	price_usd, range_km, max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr	Scatter plot; comparison bars
Customer and safety	customer_rating, safety_rating_stars, manufacturer, category	Bar chart; rating comparison
Geography and time	manufacturing_country, market_region, model_year, annual_sales_units	Bar chart; year trend
Sustainability/use case	co2_saved_tons_per_year, recommended_use_case	Bar chart; use-case comparison

Assumptions and Limitations

- Check missing values, duplicates, data types and unusual values before building final visuals.
- Confirm the meaning and units of annual_sales_units and all measures before interpreting totals or averages.
- Use consistent aggregations: for example, SUM for recorded sales and AVG for measures such as range, price or rating where appropriate.
- Do not assume that the dataset represents every EV sold globally; conclusions apply to the supplied dataset.
- Do not claim that price causes range or that ratings prove overall quality; describe observed patterns only.